

A finite-entropy separation of microstates and nonmicrostates free entropy

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Abstract

We answer the finite-entropy equality question for microstates and nonmicrostates free entropy negatively. We construct a bounded self-adjoint tuple X in a von Neumann algebra with faithful normal tracial state such that $-\infty < \chi(X) \leq \chi^*(X) - \frac{1}{2} < \infty$. Here χ is the original microstates entropy with an operator-norm cutoff and a limsup over matrix sizes. The counterexample uses a large but fixed number of variables.

1 Introduction

Free entropy connects the joint moments of noncommuting variables with the geometry of their matrix approximations. For a bounded self-adjoint tuple $X = (X_1, \dots, X_n)$ in a tracial von Neumann algebra, Voiculescu's microstates free entropy $\chi(X)$ measures the asymptotic volumes of matrix tuples whose normalized trace moments approximate those of X [12]. His nonmicrostates free entropy $\chi^*(X)$ instead uses conjugate variables, the free analogues of classical scores, and integrates free Fisher information along addition of freely independent semicircular noise [13]. We give both definitions, with their normalizations, in Section 2.

The two quantities agree for a single self-adjoint variable, and their agreement for general tuples was one of the questions in Voiculescu's unification problem [14, Sections 3.5 and 3.8]. The finite-entropy formulation asks whether

$$\chi(X) > -\infty \implies \chi(X) = \chi^*(X).$$

Guionnet stated this formulation explicitly [4, Conjecture 7.1], using the equivalent Gaussian-relative normalization. The finiteness hypothesis is substantial: as discussed by Jekel and Pi [6, Introduction], there are nonapproximable laws with $\chi = -\infty < \chi^*$. Here we construct a strict gap with both entropies finite.

Biane, Capitaine, and Guionnet proved the general comparison $\chi \leq \chi^*$ through large-deviation bounds for matrix Brownian motion [1]. Jekel and Pi obtained an elementary proof using classical entropy and Fisher information, and extended the comparison to conditional free entropy [6]. Equality is known in significant regular convex settings. Dabrowski proved it for a class of strictly convex potentials with Lipschitz derivatives and noncommutative polynomial approximations [2, Theorem A]; Jekel established entropy and Fisher-information convergence for convex matrix potentials with appropriate regularity, concentration, and trace-polynomial approximation [5, Theorem 7.1]. These results also identify conditions under which the upper and lower matrix-size limits agree. Our construction shows that finite entropy alone does not imply equality.

Theorem 1.1. *There exist an integer $n \geq 2$ and a bounded self-adjoint tuple $X = (X_1, \dots, X_n)$ in a von Neumann algebra with faithful normal tracial state such that*

$$-\infty < \chi(X) \leq \chi^*(X) - \frac{1}{2} < \infty.$$

Here χ is the original microstates entropy using an operator-norm cutoff and a limsup over matrix sizes.

Thus Theorem 1.1 answers the finite-microstates-entropy equality question negatively. The construction uses a large but fixed number of variables. It does not address equality for every smaller prescribed number of variables.

The proof compares two noise scales. Start with $C = (B_1, \dots, B_p, Y)$, where $B_1, \dots, B_p$ are free variance-one semicircular variables and Y , in a separate tensor factor, is uniformly distributed on p equally spaced points in $[-1, 1]$. Thus Y commutes with every B_i . Let S be a standard semicircular $(p+1)$ -tuple free from C . A small perturbation $X = C + \sqrt{h}S$ has finite microstates entropy, while its last coordinate still determines spectral blocks containing much of the energy of the first p coordinates.

After an additional noise time t , we obtain two different estimates. The analytic estimate uses Speicher's noncrossing cumulant calculus [9, 7] and orthogonality of semicircular Wick polynomials. Proposition 3.2 bounds the Fisher deficit by squared ℓ^2 norms of the higher cumulant arrays. The sparsity of the signal's cumulants then makes the nonmicrostates entropy of $C + \sqrt{h+t}S$ close to the entropy of the semicircular family with the same coordinate variances. The matrix estimate shows that uniform laws on a selected sequence of increasingly accurate cutoff microstate sets of X , evolved by independent GUE noise of variance t , retain detectable simultaneous block structure. This event has probability bounded away from zero uniformly along the selected matrix sizes for those ensembles, and probability at most e^{-cd^2} for size- d independent GUE matrices with the coordinate variances of the limiting law, for a fixed $c > 0$. Relative-entropy data processing therefore gives a deficit bounded away from zero in normalized classical entropy. The use of commutation constraints and projection geometry to control microstate volumes has antecedents in Ge's argument for prime factors [3, Lemma 2 and its proof outline]. The unitary-net estimate below uses the elementary volumetric covering method discussed by Szarek [10]. The event estimated here concerns the persistence of block energy after positive Gaussian time. The analytic entropy deficit becomes small when $t/\sqrt{p} \rightarrow \infty$, while the Gaussian block estimate still beats the unitary covering cost when $t \log p/p \rightarrow 0$. The choice $t = p^{3/4}$ satisfies both conditions as $p \rightarrow \infty$. We first choose one sufficiently large finite p and then h ; all matrix limits are taken with these parameters fixed.

The transfer between the scales uses the classical-to-free Fisher comparison of Jekel and Pi [6, Proposition 6.1]. Lemma 5.1 proves the needed finite-time statement while retaining the entropies of this same sequence of terminal laws. The limsup of those normalized entropies need not equal the microstates entropy of the limiting law. Keeping this sequence in the comparison permits the positive-time structural deficit to produce the strict gap for X .

The companion paper *An isomorphism of the free group factors* [8, Corollary 7.1] proves that each of the four free entropy dimensions $\delta, \delta_0, \delta^*, \delta^*$ attains every integer value $n \geq 2$ on suitable finite self-adjoint W^* -generating tuples of $L(\mathbb{F}_2)$. That statement varies the generating tuple of a fixed algebra; Theorem 1.1 compares χ and χ^* on one tuple, with both values finite.

Section 2 fixes the conventions and establishes the matrix convergence facts, starting from Gaussian asymptotic freeness [11, 7]. Section 3 proves the cumulant/Fisher estimate. Section 4 constructs the signal, fixes the two noise scales, and proves $\chi(X) > -\infty$. Section 5 proves the specified-ensemble entropy-production lemma. Section 6 establishes the block event and combines the estimates to prove Theorem 1.1.

2 Conventions and preliminary facts

We fix the entropy normalizations and matrix convergence facts used at both noise scales. The matrix inputs may be random tuples supported on bounded microstate sets.

All noncommutative variables below belong to tracial von Neumann algebras with faithful normal tracial state τ . Write $\|a\|_2^2 = \tau(a^*a)$. A standard semicircular family has variance one in each coordinate and is free. Every noise family is free from the tuple it perturbs.

For a bounded self-adjoint n -tuple X , let $\Gamma_R(X; m, d, \varepsilon)$ consist of self-adjoint $d \times d$ matrix tuples of coordinate operator norms at most R whose normalized traces of all words of length at most m differ from those of X by less than ε . With volume taken in the real Euclidean space defined by unnormalized Hilbert–Schmidt inner product, set

$$\chi_R(X) = \inf_{m, \varepsilon} \limsup_{d \rightarrow \infty} \left[d^{-2} \log \text{Vol } \Gamma_R(X; m, d, \varepsilon) + \frac{n}{2} \log d \right], \quad \chi(X) = \sup_R \chi_R(X).$$

For a tuple Z , its conjugate variables $\xi_i \in L^2(W^*(Z), \tau)$ satisfy

$$\tau(\xi_i P(Z)) = (\tau \otimes \tau)((\partial_i P)(Z)),$$

where P is a noncommutative polynomial in formal indeterminates $x_1, \dots, x_n$, and the difference quotient satisfies $\partial_i x_j = \delta_{ij} 1 \otimes 1$ and obeys the Leibniz rule. Put $\Phi^*(Z) = \sum_i \|\xi_i\|_2^2$, or infinity if they do not exist. We will use the following form of the conjugate-variable definition.

Lemma 2.1 (Polynomial variational formula). *If a bounded self-adjoint tuple Z has conjugate variables, then*

$$\Phi^*(Z) = \sup_{P_1=P_1^*, \dots, P_n=P_n^*} \sum_{i=1}^n \left[2(\tau \otimes \tau)((\partial_i P_i)(Z)) - \tau(P_i(Z)^2) \right], \quad (1)$$

where the supremum is over self-adjoint noncommutative polynomials. It suffices to take $P_i = (Q_i + Q_i^*)/2$ with coefficients of Q_i in $\mathbb{Q} + i\mathbb{Q}$. Consequently, along a family with continuous moments and existing conjugate variables, free Fisher information is a measurable function of the parameter.

Proof. The identity $\partial_i(P^*) = \text{flip}((\partial_i P)^*)$ and the symmetry of $\tau \otimes \tau$ show that ξ_i^* is a conjugate variable whenever ξ_i is. Uniqueness follows from polynomial L^2 density, so ξ_i is self-adjoint. The expression under the supremum in (1) equals

$$\sum_i (2\tau(\xi_i P_i(Z)) - \|P_i(Z)\|_2^2) = \sum_i (\|\xi_i\|_2^2 - \|\xi_i - P_i(Z)\|_2^2).$$

For completeness, the closed span of polynomial vectors is invariant under each bounded self-adjoint left multiplication operator Z_j . Its orthogonal projection therefore commutes with these operators and with $W^*(Z)$; since it fixes 1, it fixes every bounded algebra vector. Those vectors are dense in L^2 , proving polynomial density. Symmetrizing the approximants proves the claimed supremum. Approximating their finitely many coefficients proves the countable version. Each functional in that countable supremum is continuous in the moments, which proves measurability. $\square$

Our nonmicrostates normalization is

$$\chi^*(Z) = \frac{n}{2} \log(2\pi e) + \frac{1}{2} \int_0^\infty \left[\frac{n}{1+s} - \Phi^*(Z + \sqrt{s}S) \right] ds. \quad (2)$$

Write $\text{tr}_d = d^{-1} \text{Tr}$ and $\|D\|_{2,d}^2 = \sum_i \text{tr}_d(D_i^* D_i)$ for matrix tuples. A standard GUE tuple $G^{(d)}$ consists of independent centered Gaussian matrices whose real Hilbert–Schmidt coordinates have covariance $1/d$. For a random matrix tuple with density f , its normalized classical entropy is

$$H_d = -\frac{1}{d^2} \int f \log f + \frac{n}{2} \log d.$$

Thus independent GUE coordinates of variances $w_i > 0$ have normalized entropy $\frac{1}{2} \sum_i \log(2\pi e w_i)$.

Voiculescu’s asymptotic-freeness theorem introduced the random-matrix input used here [11]. We use its arbitrary deterministic-tuple form in [7, Theorem 22.35]: if deterministic matrix tuples converge in normalized trace moments, adjoining independent GUE matrices gives convergence of expected normalized traces to those of freely adjoined standard semicirculars. The following lemma supplies the additional probability and expectation statements needed in this paper.

Lemma 2.2 (Bounded matrix inputs). *Let K_d be sets of self-adjoint matrix tuples with a common coordinate operator-norm bound. Suppose their normalized trace moments converge uniformly over K_d to the moments of a bounded tuple T . Let $A^{(d)}$ be any random tuple supported on K_d , independent of a fixed finite standard GUE tuple $G^{(d)}$. Then every normalized polynomial trace in $(A^{(d)}, G^{(d)})$ converges in probability to its value at (T, S) , where S is a standard semicircular family free from T . Products of finitely many such traces converge in expectation as well. The dimensions may range over any sequence tending to infinity.*

Proof. We first record bounds independent of the deterministic input. A $1/4$ -net of the unit sphere of $\mathbb{C}^d$, viewed in real dimension $2d$, has cardinality at most 9^{2d} . For a Hermitian matrix G , $\|G\| \leq 2 \max_v |v^* G v|$ over this net. For standard GUE, each $v^* G v$ is a real centered Gaussian of variance $1/d$, since $\text{Tr}((vv^*)^2) = 1$. Hence

$$\mathbb{P}(\|G\| > L) \leq 2 \exp(d(2 \log 9 - L^2/8)). \quad (3)$$

In particular, the coordinate norms of any fixed finite GUE tuple are at most 8 with probability tending to one. For sufficiently large L the right side of (3) is at most $2 \exp(-dL^2/16)$. Integrating that tail shows that every fixed moment of the operator norm is bounded uniformly in d .

Now take a deterministic input a with coordinate norms at most R , and put $F = \text{tr}_d P(a, G)$ for a fixed polynomial P . Let (E_α) be a Hermitian basis orthonormal for the unnormalized Hilbert–Schmidt inner product, and write $G_j = \sum_\alpha g_{j\alpha} E_\alpha$. The real coordinates $g_{j\alpha}$ are independent Gaussians with variance $1/d$. Cyclicity of trace gives

$$\partial_{g_{j\alpha}} F = d^{-1} \text{Tr}(E_\alpha B_j), \quad \sum_{j,\alpha} |\partial_{g_{j\alpha}} F|^2 = d^{-1} \sum_j \text{tr}_d(B_j^* B_j),$$

where B_j is the sum of the cyclically ordered products obtained by deleting each occurrence of the j th GUE letter in P . The basis also complexifies to an orthonormal basis of all matrices, so the second identity applies to complex polynomials.

For a polynomial in standard real Gaussian coordinates, expansion in orthonormal Hermite polynomials shows that its variance is at most the expected squared gradient: differentiation weights each nonconstant coefficient’s squared modulus by its positive total degree. Rescaling the coordinates to variance $1/d$, and applying this to real and imaginary parts, gives

$$\mathbb{E}|F - \mathbb{E}F|^2 \leq d^{-2} \sum_j \mathbb{E} \text{tr}_d(B_j^* B_j) = O_{P,R}(d^{-2}). \quad (4)$$

The final bound follows from the norm-moment estimate, since each B_j has fixed degree and the deterministic coordinates are uniformly bounded. Thus the cited expected-trace theorem yields convergence in probability for each deterministic moment-convergent input sequence.

This convergence is uniform over the sets K_d . Otherwise, for some polynomial and positive error threshold, one could choose a subsequence and points $a_d \in K_d$ whose error probabilities stay bounded away from zero. These points form a bounded deterministic moment-convergent sequence, contradicting the preceding conclusion. Conditioning on $A^{(d)}$ proves convergence in probability for the random input.

Finally, any fixed product Q of normalized polynomial traces satisfies

$$|Q(A^{(d)}, G^{(d)})| \leq C(1 + \max_j \|G_j^{(d)}\|)^D$$

for constants C, D depending only on the polynomials and the common cutoff. The same estimate squared has uniformly bounded expectation. The products are therefore uniformly integrable, so their convergence in probability implies convergence of their expectations. $\square$

3 A cumulant bound for the Fisher deficit

We first estimate how far the Fisher information of a semicircularly perturbed tuple is from that of a semicircular family with the same covariance. The estimate will use the squared sizes of the higher cumulant arrays, which will allow us to exploit the sparse cumulants of the signal in the next section.

For a centered bounded self-adjoint n -tuple C , write $\kappa_m(C)$ for its order- m free-cumulant array, with

$$\|\kappa_m(C)\|_2^2 = \sum_{i_1, \dots, i_m=1}^n |\kappa_m(C_{i_1}, \dots, C_{i_m})|^2.$$

We use the moment-cumulant formula and its Möbius inversion on noncrossing partitions, the vanishing of mixed cumulants between free algebras, and the fact that a standard semicircular family has only second cumulants, given by its covariance [7, Proposition 11.4, Theorem 11.16, and Examples 11.21].

The following testing identity converts the higher-cumulant terms into an orthogonal series. For a bounded self-adjoint tuple Z , write $E[\cdot \mid W^*(Z)]$ for the trace-preserving conditional expectation, also viewed as the orthogonal projection onto $L^2(W^*(Z), \tau)$ when its argument belongs only to L^2 .

Lemma 3.1 (Wick testing). *Let C be a centered bounded self-adjoint n -tuple, let S be a standard semicircular family free from C , and put $Z = C + \sqrt{u}S$, where $u > 0$. For a multi-index $\mathbf{j} = (j_1, \dots, j_r)$, $r \geq 1$, and a monomial P in the indeterminates $x_1, \dots, x_n$, define*

$$D_{\mathbf{j}}(P) = \sum_{P=P_0 x_{j_1} P_1 \dots x_{j_r} P_r} \prod_{a=0}^r \tau(P_a(Z)).$$

The sum selects increasing positions with the prescribed indices; empty gap words P_a contribute one. Extend $D_{\mathbf{j}}$ linearly to polynomials. There is an orthonormal family of semicircular Wick polynomials $U_{\mathbf{j}}(S)$, indexed by all such multi-indices, with

$$D_{\mathbf{j}}(P) = u^{-r/2} \tau(U_{\mathbf{j}}(S)^* P(Z)). \quad (5)$$

In particular, the conjugate variables of Z exist and are

$$\xi_i = u^{-1/2} E[S_i \mid W^*(Z)].$$

Proof. Define $U_{\mathbf{j}}$ from the word $S_{j_1} \cdots S_{j_r}$ by summing over sets of disjoint adjacent pairs in the *original* index list: delete each chosen pair with factor $-\delta_{j_a, j_{a+1}}$. The empty set gives the original word. This definition also gives $U_{\mathbf{j}}^* = U_{(j_r, \dots, j_1)}$.

Expand $\tau(U_{\mathbf{j}}^* P(Z))$ by noncrossing cumulants. After multilinearly expanding $Z = C + \sqrt{u}S$, freeness and semicircularity show that every block meeting an initial S position is a pair. Its covariance is δ_{ij} against another initial S and $\sqrt{u}\delta_{ij}$ against a selected S term from Z .

To account for the deletion terms, fix a choice of C or S at every P -position in the multilinear expansion, and a noncrossing partition π on the original positions. Let $A(\pi)$ be its initial-initial pair blocks whose endpoints are originally adjacent. For these fixed choices, contributions correspond exactly to subsets $F \subseteq A(\pi)$: delete the pairs in F and restore them after expanding the remaining word. Restoring adjacent pairs preserves noncrossingness and the covariance factors, so their total coefficient is

$$\sum_{F \subseteq A(\pi)} (-1)^{|F|} = (1 - 1)^{|A(\pi)|}.$$

If π has any initial-initial pair, an innermost one is originally adjacent. Indeed, its intervening positions are all initial, and each must pair inside that interval to avoid crossing. Thus $A(\pi)$ is nonempty and the partition cancels. This argument includes nested internal pairs and does not require deleting newly adjacent positions.

In each surviving partition all initial positions pair externally with positions of P , in reverse order. The initial indices in $U_{\mathbf{j}}^*$ are $j_r, \dots, j_1$, so the selected indices in P occur as $j_1, \dots, j_r$. The external pair arches prevent any remaining block from connecting distinct gaps. Summing the unrestricted partitions and the choices of C or S within each gap therefore gives its trace, even though the entries of C need not be free. Each external pair contributes $\sqrt{u}$, proving (5).

Apply this calculation with zero signal and $u = 1$. A Wick word tests to zero against a shorter word and gives the corresponding Kronecker delta against a word of its own degree. Since every other term of a Wick polynomial has smaller degree, this proves orthonormality against Wick words of equal or smaller degree; conjugate symmetry covers the remaining cases. For $r = 1$, $D_i(P) = (\tau \otimes \tau)(\partial_i P(Z))$, so (5) and conditional expectation give the stated conjugate variables. $\square$

Proposition 3.2 (Squared-cumulant Fisher bound). *Let C be a centered bounded self-adjoint n -tuple with diagonal covariance $\tau(C_i C_j) = \delta_{ij} c_i$, where $c_i \geq 0$, and let S be a standard semicircular family free from C . Put $V_u = C + \sqrt{u}S$ and $v_i(u) = u + c_i$ for $u > 0$. For every $u > 0$,*

$$\sum_i \frac{1}{v_i(u)} \leq \Phi^*(V_u) \leq \frac{n}{u}.$$

Whenever $\sum_{r \geq 2} u^{-r} \|\kappa_{r+1}(C)\|_2^2 < \infty$,

$$0 \leq \Phi^*(V_u) - \sum_i \frac{1}{v_i(u)} \leq \frac{1}{u^2} \sum_{r \geq 2} u^{-r} \|\kappa_{r+1}(C)\|_2^2. \quad (6)$$

Proof. Let $Z = V_u$ and let ξ_i be supplied by Lemma 3.1. Contraction of conditional expectation gives $\Phi^*(Z) \leq n/u$. Also $\tau(Z_i \xi_i) = 1$ by testing the conjugate identity with Z_i . Since $\|Z_i\|_2^2 = v_i(u)$, Cauchy–Schwarz gives $\|\xi_i\|_2^2 \geq v_i(u)^{-1}$ for every $u > 0$.

For the upper deficit bound, expand the block containing the first position in $\tau(Z_i P(Z))$. Its other positions select the letters in $D_{\mathbf{j}}(P)$, while its gaps contribute their moments. Thus

$$\tau(Z_i P(Z)) = \sum_{r \geq 1, \mathbf{j}} \kappa_{r+1}(Z_i, Z_{j_1}, \dots, Z_{j_r}) D_{\mathbf{j}}(P).$$

Only $r \leq \deg P$ contributes. Centering removes the order-one cumulant, and diagonal covariance makes the order-two term $v_i(u)D_i(P)$. Adding free semicircular noise leaves all higher cumulants equal to those of C .

Define the ambient L^2 series

$$R_i = \sum_{r \geq 2, \mathbf{j}} u^{-r/2} \kappa_{r+1}(C_i, C_{j_1}, \dots, C_{j_r}) U_{\mathbf{j}}(S)^*.$$

The adjoint Wick family is orthonormal, and the assumed summability gives convergence in L^2 with

$$\sum_i \|R_i\|_2^2 = \sum_{r \geq 2} u^{-r} \|\kappa_{r+1}(C)\|_2^2.$$

For a fixed polynomial P , all terms with $r > \deg P$ have zero trace pairing by (5). Moreover, $A \mapsto \tau(AP(Z))$ is continuous on L^2 . The finite cumulant expansion above therefore gives

$$\tau((Z_i - v_i(u)\xi_i)P(Z)) = \tau(R_i P(Z)).$$

The coefficients in R_i need no complex conjugation: this identity uses the bilinear trace pairing, rather than a Hilbert inner product. Testing with adjoint polynomials converts it into Hilbert-space orthogonality. Polynomial L^2 density then identifies

$$Z_i - v_i(u)\xi_i = E[R_i \mid W^*(Z)].$$

The conjugate variables are self-adjoint by their conditional-expectation formula. Expanding the residual norm using $\tau(Z_i \xi_i) = 1$ yields

$$\|Z_i - v_i(u)\xi_i\|_2^2 = v_i(u)^2 (\|\xi_i\|_2^2 - v_i(u)^{-1}).$$

Summing after division by $v_i(u)^2$, using projection contraction and $v_i(u) \geq u$, proves (6). $\square$

The proposition reduces the large-noise entropy estimate to control of the cumulant arrays of the signal. Its separate bound $\Phi^*(V_u) \leq n/u$ will also ensure integrability on every compact interval of positive noise times.

4 The signal and its two noise scales

The signal must satisfy two competing requirements. Its higher cumulants must be sparse enough for the Fisher estimate, while many coordinates must share a block decomposition. Tensor independence supplies the latter without spoiling the former.

Fix an integer $p \geq 2$ and set $n = p + 1$. Let $B_1, \dots, B_p$ be free variance-one semicirculars. In a separate tensor factor take a variable Y with uniform distribution on p equally spaced points $y_l = -1 + 2(l-1)/(p-1)$, $1 \leq l \leq p$. Thus Y commutes with every B_i , but is not being taken free from them. Set

$$C = (B_1, \dots, B_p, Y), \quad V_a = C + \sqrt{a}S, \quad L(a) = \frac{1}{2} \sum_{i=1}^n \log(2\pi e(a + c_i)),$$

where $c_i = \tau(C_i^2)$. Thus $c_i = 1$ for $i \leq p$, and C is centered with diagonal covariance. The benchmark $L(a)$ is the nonmicrostates entropy of the centered semicircular family with coordinate variances $a + c_i$, and also the normalized classical entropy of the corresponding independent GUE law.

The next construction provides a faithful normal tracial realization together with uniformly bounded deterministic matrix models for C . These models will later give the finite lower bound for $\chi(X)$. Choose uniformly bounded deterministic matrix models $B^{(q_k)}$ of the free semicircular p -tuple, using the convergence and norm bounds in Section 2. Tensor them with I_p and take $I_{q_k} \otimes \text{diag}(y_1, \dots, y_p)$ for Y . In dimensions $d_k = pq_k$ these give models of the required tensor law. Adjoin an independent GUE n -tuple and select realizations satisfying successively more moment tests and one fixed norm bound. Their joint limiting moments give a positive tracial functional in the variables (C, S) with the asserted freeness. In its GNS construction, left and right coordinate multiplication are bounded: the matrix inequality $\text{tr}_d(Q^* D_i^2 Q) \leq R_i^2 \text{tr}_d(Q^* Q)$ passes to the limit. The vacuum is cyclic for right polynomial multiplication, which commutes with the weak closure of the left algebra, so the vacuum is separating for that closure. Its vector state is therefore faithful and normal; the trace identity extends by separate weak continuity. This supplies the required tracial von Neumann algebra. Further independent noise families can be included in the same construction. Faithfulness and the semicircular distribution give $\|B_i\| = \|S_i\| = 2$.

Lemma 4.1. *For $K = 128$ and every $m \geq 1$,*

$$\|\kappa_m(C)\|_2 \leq K^m p^{m/4}.$$

Proof. There are at most 4^m noncrossing partitions and their Möbius coefficients have magnitude at most 4^m , by the Catalan product formula and the interval factorization of the noncrossing lattice [7, Proposition 9.4, Theorem 9.29, and Propositions 10.14–10.15]. Each product of block moments has absolute value at most 2^m , since $\|B_i\| = 2$ and $\|Y\| \leq 1$. Thus each cumulant is at most 32^m in absolute value.

Suppose b of the m positions have B indices. A nonzero product of block moments in Möbius inversion requires, after deleting the Y positions in each block, a semicircular word with a noncrossing pairing matching equal indices. Indeed, tensor independence factors each block moment into its B -word moment and its Y -moment. The union of these pairings is still noncrossing: crossing pairs from distinct blocks would cross the containing partition. For fixed b positions, at most $4^b p^{b/2}$ labelings can therefore contribute. Summing over positions bounds the number of potentially nonzero entries by $(1 + 4\sqrt{p})^m$. Consequently

$$\|\kappa_m(C)\|_2 \leq 32^m (1 + 4\sqrt{p})^{m/2} \leq (32\sqrt{5})^m p^{m/4} \leq K^m p^{m/4}.$$

□

By Proposition 3.2 and a geometric series with ratio $K^2\sqrt{p}/u < 1/2$, for $u > 2K^2\sqrt{p}$ we have

$$0 \leq \Phi^*(V_u) - \sum_i \frac{1}{u + c_i} \leq 2K^6 p^{3/2} u^{-4}. \quad (7)$$

Adding independent free semicircular noise of variance s to V_a gives the law of V_{a+s} . Equation (2) therefore yields

$$\chi^*(V_a) = L(a) - \frac{1}{2} \int_a^\infty \left[\Phi^*(V_u) - \sum_i \frac{1}{u + c_i} \right] du. \quad (8)$$

All terms are finite for $a > 0$: the integrand is measurable by (1), nonnegative, and bounded above by

$$\frac{n}{u} - \sum_i \frac{1}{u + c_i} = \sum_i \frac{c_i}{u(u + c_i)} \leq \frac{\sum_i c_i}{u^2},$$

which is integrable on $[a, \infty)$. In deriving (8), the covariance correction is $\int_0^\infty [(1+s)^{-1} - (a+c_i+s)^{-1}] ds = \log(a+c_i)$; thus no subtraction of divergent integrals is used. In particular,

$$\chi^*(V_a) \geq L(a) - \frac{K^6 p^{3/2}}{3a^3} > L(a) - \frac{1}{2} \quad (a > 2K^2 \sqrt{p}), \quad (9)$$

$$\chi^*(V_{h+t}) - \chi^*(V_h) = \frac{1}{2} \int_0^t \Phi^*(V_{h+s}) ds. \quad (10)$$

We now fix p sufficiently large and put $t = p^{3/4}$ so that

$$\begin{aligned} t &> 2K^2 \sqrt{p}, & \frac{32t}{p} &< \frac{1}{8}, \\ \frac{p}{256(2+t)} &> 2 \log(1 + 128\sqrt{2+t}) + \log 2 + 8. \end{aligned} \quad (11)$$

These inequalities are compatible because $p/t = p^{1/4}$ grows faster than $\log p$. An explicit choice is $p = 2^{64}$ and $t = 2^{48}$: the first threshold is 2^{47} and $32t/p = 2^{-11}$. For the last inequality, its left side exceeds 128 since $2+t < 2^{49}$, whereas its right side is less than $67 \log 2 + 8 < 75$ since $1 + 128\sqrt{2+t} < 2^{33}$. From this point on, p and t are fixed, and only matrix sizes will tend to infinity.

For small $h > 0$, partition the spectrum of $X_n = Y + \sqrt{h}S_n$ at the midpoints between the y_l and denote its spectral projections by $e_l(h)$. Let $\Delta = 2/(p-1)$ be the atom spacing. Since $\|\sqrt{h}S_n\| = 2\sqrt{h}$, the condition $h < \Delta^2/64$ places the spectrum inside the disjoint $\Delta/4$ -neighborhoods of the atoms, with every threshold in a spectral gap. Choose continuous functions agreeing with the interval indicators on these common neighborhoods. Continuous functional calculus, or uniform polynomial approximation there followed by norm continuity, gives $e_l(h) \rightarrow e_l(0)$ in norm. At $h = 0$ these projections commute with every B_i , have trace $1/p$, and satisfy

$$\sum_{i=1}^p \sum_{l=1}^p \|e_l(0)B_i e_l(0)\|_2^2 = \sum_{i=1}^p \|B_i\|_2^2 = p.$$

We may therefore fix $0 < h < \min\{1, \Delta^2/64\}$ and abbreviate $e_l = e_l(h)$ so that, for $X = V_h$,

$$\tau(e_l) < \frac{3}{2p} \quad (1 \leq l \leq p), \quad \sum_{i=1}^p \sum_{l=1}^p \|e_l X_i e_l\|_2^2 > \frac{3p}{4}. \quad (12)$$

Lemma 4.2. *The tuple X has $\chi(X) > -\infty$.*

Proof. Use the deterministic models for C above, in dimensions $d_k = pq_k$, and add independent GUE noise of variance h . If M bounds the coordinate norms of those models, take $R_0 = M + 8\sqrt{h}$. Lemma 2.2 and (3) show that these noised models lie in every fixed moment neighborhood of X , within that cutoff, with probability tending to one. Their density is bounded above by $(d/(2\pi h))^{nd^2/2}$. If q_d is the indicated probability, then

$$\text{Vol } \Gamma_{R_0}(X; m, d, \varepsilon) \geq q_d (2\pi h/d)^{nd^2/2}.$$

Since $q_d \rightarrow 1$ along the model dimensions, the normalized limsup logarithm is at least $\frac{n}{2} \log(2\pi h)$ for each m, ε . Take the infimum and then the supremum over cutoffs. The limsup definition makes this subsequence of matrix sizes sufficient. $\square$

5 Entropy production for specified matrix ensembles

We use the classical-to-free Fisher comparison of Jekel and Pi [6, Proposition 6.1], keeping the terminal ensemble in the conclusion. The purpose is to transfer a later entropy bound for one evolved matrix law back to the initial microstates entropy. We prove the comparison for the exact laws that will enter the structural event in the next section.

Lemma 5.1. *For the fixed tuple $X = V_h$, fix a cutoff $R > 0$ with $\chi_R(X) > -\infty$. There are dimensions tending to infinity and random tuples $A^{(d)}$, uniform on increasingly accurate positive-volume microstate sets at cutoff R , such that, for independent standard GUE tuples,*

$$\limsup_d H_d(A^{(d)} + \sqrt{t}G^{(d)}) \geq \chi_R(X) + \frac{1}{2} \int_0^t \Phi^*(V_{h+s}) ds. \quad (13)$$

The left side is the limsup of the normalized entropies of these particular ensembles.

Proof. Every uniform law on a positive-volume microstate set at cutoff R has $\sum_i \mathbb{E} \operatorname{tr}_d(A_i^2) \leq nR^2$. Comparison with independent GUE matrices of variance R^2 bounds its normalized entropy above by $\frac{n}{2} \log(2\pi e R^2)$. Thus $\chi_R(X)$ is a finite real number. Choose a new increasing sequence of dimensions d_k , $k \geq 1$, such that the normalized log-volume at parameters $(m, \varepsilon) = (k, 1/k)$ is at least $\chi_R(X) - 1/k$. Such arbitrarily large dimensions exist by the defining limsup, and the selected sets have positive volume. The pairs $(k, 1/k)$ are cofinal among all moment accuracies. Their uniform laws satisfy

$$\liminf_d H_d(A^{(d)}) \geq \chi_R(X). \quad (14)$$

Here and below d ranges over the selected dimensions d_k . Every tuple in the support of $A^{(d)}$ has the same fixed cutoff and increasingly accurate moments of X . The selection does not depend on the subsequent noise time or polynomial tests.

Put $Z^{(d)}(s) = A^{(d)} + \sqrt{s}G^{(d)}$, and let f_s be its density for $s > 0$. In the nd^2 real Hilbert–Schmidt coordinates, the added Gaussian has covariance $(s/d)I$. Thus $\partial_s f_s = (1/(2d))\Delta f_s$. Differentiating the entropy and integrating by parts gives the normalized de Bruijn identity

$$\frac{d}{ds} H_d(Z^{(d)}(s)) = \frac{1}{2} I_d(s), \quad I_d(s) = \frac{1}{d^3} \int \|\nabla \log f_s\|^2 f_s = \sum_i \mathbb{E} \operatorname{tr}_d(F_i^2), \quad (15)$$

where $F_i = -(1/d)\nabla_i \log f_s$ evaluated at $Z^{(d)}(s)$. Gaussian convolution of a compactly supported distribution justifies differentiation and integration by parts on compact positive-time intervals. Indeed, derivatives have Gaussian decay, a Gaussian lower bound on f_s bounds $|\log f_s(x)|$ by $C(1 + \|x\|^2)$, and

$$\nabla \log f_s(x) = -\frac{d}{s}(x - \mathbb{E}[A^{(d)} \mid Z^{(d)}(s) = x])$$

has at most linear growth. Here the norm and gradient are Euclidean, and the conditional expectation is bounded at each fixed dimension. In particular the entropy and Fisher information are finite at positive times. Entropy concavity under mixtures of translations gives $H_d(Z^{(d)}(\varepsilon)) \geq H_d(A^{(d)})$. Integrating first from ε to t and then letting $\varepsilon \downarrow 0$ proves

$$H_d(Z^{(d)}(t)) \geq H_d(A^{(d)}) + \frac{1}{2} \int_0^t I_d(s) ds. \quad (16)$$

No continuity of entropy at zero is needed.

For fixed $s > 0$, integration by parts against any complex noncommutative polynomial P gives

$$\mathbb{E} \operatorname{tr}_d(F_i P(Z^{(d)}(s))) = \mathbb{E}(\operatorname{tr}_d \otimes \operatorname{tr}_d)((\partial_i P)(Z^{(d)}(s))). \quad (17)$$

To verify this identity including its normalization, take an orthonormal Hermitian basis (E_α) for unnormalized Hilbert–Schmidt inner product. It satisfies $\sum_\alpha E_\alpha D E_\alpha = \operatorname{Tr}(D)1$ for every complex matrix D . Scalar integration by parts, separately on real and imaginary parts, turns the left side of (17) into

$$d^{-2} \sum_\alpha \mathbb{E} \operatorname{Tr}(E_\alpha D_i P(Z^{(d)}(s)) [E_\alpha]).$$

Here $D_i P(Z)[E_\alpha]$ is the directional derivative in the i th matrix coordinate along E_α . At an occurrence $P = LZ_i R$ in a monomial, its contribution is

$$d^{-2} \sum_\alpha \operatorname{Tr}(E_\alpha L E_\alpha R) = d^{-2} \operatorname{Tr}(L) \operatorname{Tr}(R) = \operatorname{tr}_d(L) \operatorname{tr}_d(R).$$

Summing over occurrences proves (17) with the prefix and suffix in the same order as the free difference quotient.

For self-adjoint polynomial tests P_i , completion of squares gives

$$I_d(s) \geq \sum_i \mathbb{E} \operatorname{tr}_d (2F_i P_i(Z^{(d)}(s)) - P_i(Z^{(d)}(s))^2).$$

First use (17) to replace every term involving the score by a product of normalized polynomial traces. Lemma 2.2 then gives the limit of the right side as

$$\sum_i [2(\tau \otimes \tau)((\partial_i P_i)(V_{h+s})) - \tau(P_i(V_{h+s})^2)].$$

The limiting law is V_{h+s} because the new semicircular noise is free from $X = V_h$. Conjugate variables exist by Lemma 3.1. Taking the supremum over these tests in Lemma 2.1 gives

$$\liminf_d I_d(s) \geq \Phi^*(V_{h+s}). \quad (18)$$

This holds for every fixed $s > 0$ along the same selected dimensions. The moments of $V_{h+s} = C + \sqrt{h+s}S$ are continuous in s , so Lemma 2.1 also gives measurability of the free Fisher information. Applying Fatou’s lemma to the nonnegative measurable functions I_d , and then using (14) and (16), yields

$$\liminf_d H_d(Z^{(d)}(t)) \geq \chi_R(X) + \frac{1}{2} \int_0^t \Phi^*(V_{h+s}) ds.$$

This proves (13), even with $\liminf$ in place of $\limsup$. The free Fisher integral is finite since $\Phi^*(V_{h+s}) \leq n/(h+s)$ and $h > 0$. Throughout the proof the terminal law is that of the same $A^{(d)} + \sqrt{t}G^{(d)}$ selected above; no entropy-maximizing law at the terminal distribution has been chosen. $\square$

6 A persistent block-structure entropy deficit

We now bound the entropy of the particular terminal ensembles retained in Lemma 5.1. Fix its cutoff R , dimension subsequence, and uniform microstate ensembles $A^{(d)}$. The parameters p, h, t

remain fixed as in Section 4. For a decomposition of the identity into p orthogonal projections $P = (P_l)$, write

$$\mathcal{D}_P D = \left(\sum_l P_l D_l P_l \right)_i, \quad D_{[p]} = (D_1, \dots, D_p).$$

Use the midpoint spectral partition of the last coordinate of $A^{(d)}$ to obtain P . For every tuple in its support and all sufficiently large dimensions,

$$\text{rank}(P_l) \leq \frac{2d}{p}, \quad \|\mathcal{D}_P A_{[p]}^{(d)}\|_{2,d}^2 > p/2. \quad (19)$$

To prove this uniformly, first consider any deterministic sequence of tuples at cutoff R whose joint moments converge to those of X . Write I_l for the midpoint intervals, with any fixed convention at their endpoints, and let ν_d be the empirical spectral measure of $A_n^{(d)}$. These measures converge weakly to the law of X_n . Choose a union N of small closed neighborhoods of the midpoint thresholds, disjoint from $\sigma(X_n)$. On a common compact interval containing $[-R, R]$ and $\sigma(X_n)$, choose continuous functions $0 \leq f_l \leq 1$ agreeing with 1_{I_l} outside N . Then $f_l(X_n) = e_l$, and, with $F_l^{(d)} = f_l(A_n^{(d)})$,

$$\|P_l - F_l^{(d)}\|_{2,d}^2 \leq \nu_d(N) \rightarrow 0.$$

Here weak convergence gives $\nu_d(N) \rightarrow 0$ because N is closed and has zero limiting mass. In particular, small spectral outliers and eigenvalues exactly at a threshold are allowed.

The cutoff controls multiplication by every other coordinate:

$$\|P_l A_i^{(d)} P_l - F_l^{(d)} A_i^{(d)} F_l^{(d)}\|_{2,d} \leq 2R \|P_l - F_l^{(d)}\|_{2,d}. \quad (20)$$

The two compressed norms are at most R , so their squared norms also have difference tending to zero. Uniform polynomial approximation of f_l and joint moment convergence now imply

$$\frac{\text{rank}(P_l)}{d} \rightarrow \tau(e_l), \quad \|P_l A_i^{(d)} P_l\|_{2,d}^2 \rightarrow \|e_l X_i e_l\|_2^2.$$

The block summands are orthogonal in Hilbert–Schmidt inner product; hence their squared norms add to $\|\mathcal{D}_P A_{[p]}^{(d)}\|_{2,d}^2$. The strict margins in (12) give (19). If uniformity over the selected supports failed, a violating tuple in each of a subsequence of dimensions would itself be a bounded moment-convergent sequence, contradicting this argument. Closure points of the microstate sets satisfy the same moment bounds with weak inequalities, whose errors still tend to zero.

Put $v = 1 + h + t$, and let $\mathcal{E}_d$ be the following event for a matrix tuple D :

$$\begin{aligned} \|D_{[p]}\|_{2,d} &\leq 4\sqrt{vp}, & \text{there exists a decomposition } P \text{ with} \\ \text{rank}(P_l) &\leq 2d/p, & \|\mathcal{D}_P D_{[p]}\|_{2,d} \geq \sqrt{p}/4. \end{aligned} \quad (21)$$

This is a deterministic Borel event. For each admissible rank list, the space of projection decompositions is a compact unitary orbit. The maximum of $\|\mathcal{D}_P D_{[p]}\|_{2,d}$ over that orbit is a continuous function of D : every compression is contractive, so the maximum is even Lipschitz in the tuple 2-norm. The event is therefore a finite union of closed threshold sets intersected with a closed norm ball. Its definition does not depend on $A^{(d)}$; the spectral decomposition of $A_n^{(d)}$ will serve only as a witness for its probability under the evolved ensemble.

Lemma 6.1. *Let μ_d be the law of $A^{(d)} + \sqrt{t}G^{(d)}$, and let γ_d be the law of independent centered GUE coordinates of variances $v_i(h+t) = h+t+c_i$. For all sufficiently large d ,*

$$\mu_d(\mathcal{E}_d) \geq \frac{1}{2}, \quad \gamma_d(\mathcal{E}_d) \leq e^{-4d^2}.$$

Proof. For the first bound use the witness in (19), before adding the noise. Conditionally on $A^{(d)}$, its projections are fixed and independent of $G^{(d)}$. Put $r_l = \text{rank}(P_l)$. The block-diagonal Hermitian subspace has real dimension $\sum_l r_l^2$, so the GUE covariance $1/d$ and the normalized trace give

$$\mathbb{E}[\|\sqrt{t}\mathcal{D}_P G_{[p]}^{(d)}\|_{2,d}^2 \mid A^{(d)}] = tp \sum_l (\text{rank}(P_l)/d)^2 \leq 2t.$$

Markov's inequality at squared threshold $p/16$ bounds failure by $32t/p < 1/8$. On success the projected total norm exceeds $\sqrt{p/2} - \sqrt{p}/4 > \sqrt{p}/4$. Independence and centering of the noise also give

$$\mathbb{E}\|A_{[p]}^{(d)} + \sqrt{t}G_{[p]}^{(d)}\|_{2,d}^2 = \mathbb{E}\|A_{[p]}^{(d)}\|_{2,d}^2 + tp \longrightarrow vp.$$

Markov's inequality at the squared norm $16vp$ bounds the failure of the norm condition in (21) by $1/16 + o(1) < 1/8$. The union bound shows that both conditions hold with probability at least $3/4$, proving the weaker claimed bound.

For the Gaussian bound enumerate at most $(d+1)^p$ rank lists and choose an operator-norm δ -net of the unitary group, with $\delta = 1/(64\sqrt{v})$. A maximal δ -separated subset of $U(d)$ has unitary centers and is a δ -net. Its ambient operator-norm balls of radius $\delta/2$ are disjoint and lie in the ball of radius $1 + \delta/2$ in the real $2d^2$ -dimensional space $M_d(\mathbb{C})$. Volume comparison gives cardinality at most $(1 + 2/\delta)^{2d^2}$. Every decomposition of a fixed rank list is a unitary conjugate of its standard coordinate blocks P^0 , including lists with zero ranks. For unitaries U, V with $\|U - V\| \leq \delta$,

$$\|U^* D_{[p]} U - V^* D_{[p]} V\|_{2,d} \leq 2\delta \|D_{[p]}\|_{2,d}.$$

Fixed-block compression is contractive. Thus replacing a witnessing unitary by a net point loses at most $2\delta \cdot 4\sqrt{vp} = \sqrt{p}/8$ on (21). Hence some rank list satisfying the bounds and some net unitary U must satisfy

$$\|\mathcal{D}_{P^0}(U^* D_{[p]} U)\|_{2,d} \geq \sqrt{p}/8.$$

For a fixed list and unitary, unitary invariance preserves the independent variance- v GUE law of the first p coordinates. Their projected Hilbert–Schmidt vector has $q = p \sum_l \text{rank}(P_l^0)^2 \leq 2d^2$ real coordinates of variance v/d . The further division by d in the normalized squared norm gives the law $(v/d^2)Q$, where Q is chi-squared of degree q . Since $\mathbb{E}e^{Q/4} = 2^{q/2}$, its probability of reaching the squared-norm threshold $p/64$, equivalently $Q \geq pd^2/(64v)$, is at most

$$\exp \left[d^2 \left(\log 2 - \frac{p}{256v} \right) \right].$$

The full union bound consequently has logarithm at most

$$p \log(d+1) + d^2 \left[2 \log(1 + 128\sqrt{v}) + \log 2 - \frac{p}{256v} \right].$$

Since $v < 2 + t$, Equation (11) makes the bracket less than -8 . The first term is $o(d^2)$ because p is fixed. This proves the bound e^{-4d^2} for large d . $\square$

Proof of Theorem 1.1. Retain the same cutoff, dimension subsequence, and ensembles throughout. Put $\alpha = \mu_d(\mathcal{E}_d)$ and $\beta = \gamma_d(\mathcal{E}_d)$. Binary data processing for relative entropy and the bound on binary entropy give

$$\text{KL}(\mu_d \parallel \gamma_d) \geq \alpha \log \frac{\alpha}{\beta} + (1 - \alpha) \log \frac{1 - \alpha}{1 - \beta} \geq \alpha \log \frac{1}{\beta} - \log 2.$$

Thus Lemma 6.1 implies $\liminf_d d^{-2} \text{KL}(\mu_d \|\gamma_d) \geq 2$. We retain only one unit of this normalized deficit in the comparison below. At positive time the bounded-input Gaussian convolution has finite entropy and finite second moments, and its relative entropy with respect to γ_d is finite. The Gaussian reference has density

$$g_d(D) = \prod_{i=1}^n \left[\left(\frac{d}{2\pi v_i(h+t)} \right)^{d^2/2} \exp \left(-\frac{d^2 \text{tr}_d(D_i^2)}{2v_i(h+t)} \right) \right].$$

Using $-\int f \log f = -\int f \log g_d - \text{KL}(f \| g_d)$ and the normalization of H_d gives the exact identity

$$H_d(Z^{(d)}(t)) = \frac{1}{2} \sum_i \left[\log(2\pi v_i(h+t)) + \frac{\mathbb{E} \text{tr}_d(Z_i^{(d)}(t)^2)}{v_i(h+t)} \right] - d^{-2} \text{KL}(\mu_d \|\gamma_d).$$

The expected second moments equal those of $A_i^{(d)}$ plus t , and converge to $v_i(h+t)$. In particular,

$$\limsup_d H_d(Z^{(d)}(t)) \leq L(h+t) - 1.$$

This upper bound concerns exactly the terminal law $Z^{(d)}(t) = A^{(d)} + \sqrt{t}G^{(d)}$ in Lemma 5.1. Combining the two bounds on its entropy and then (9)–(10) gives

$$\begin{aligned} \chi_R(X) &\leq L(h+t) - 1 - \frac{1}{2} \int_0^t \Phi^*(V_{h+s}) ds \\ &\leq \chi^*(V_{h+t}) - \frac{1}{2} - \frac{1}{2} \int_0^t \Phi^*(V_{h+s}) ds \\ &= \chi^*(V_h) - \frac{1}{2}. \end{aligned}$$

This holds for every fixed cutoff with $\chi_R(X) > -\infty$; for the remaining cutoffs it is automatic. Taking the supremum over cutoffs preserves the bound. Lemma 4.2 gives $\chi(X) > -\infty$, and (8) shows that $\chi^*(X)$ is a finite real number. The bounded matrix realization in Section 4 supplies the asserted faithful normal tracial state and bounded self-adjoint tuple. $\square$

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